

AI STUDYBUDDY API

Solution Requirements

Functional, data, and security requirements

TEAM ID	SWTID-2026-2776
TEAM SIZE	5
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DATE	30 September 2026

Functional

- Register, login, refresh token, logout.
- Upload, list, retrieve, and delete study materials.
- Summarize material and generate flashcards, quiz, and study plan.
- Admin: list users, delete user and associated materials, return statistics.

Data constraints

User roles in the model are student/admin. Material stores user reference, title, content, filename, summary, flashcards, quiz, study plan, and timestamps. Upload accepts .txt, .md, .pdf extensions up to 5 MB.

Security requirements

JWT authentication; admin-only gates; students should access only their own materials; secrets supplied using environment variables.

Code review findings

Registration currently accepts role from request body; protect privileged role assignment. AI material handlers look up IDs but do not visibly check that a non-admin owns the material. PDF extension is allowed, but file content is read as UTF-8 rather than parsed as PDF.

Verification status

These are requirements and code observations; no completed runtime test report was provided.